

ELEC 3115

Lecture 10: Week 5, Thursday

Time-varying magnetic field and effects

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own understanding.)

Calculation of inductance inductance relationship inductance is constant
inductance and stored magnetic circuit magnetic fluxes
flux and inductance **Magnetic force** flux linkage Inductance and energy
types of inductances
Mutual inductance
stored energy **magnetic field** self inductance
magnetic energy inductance formation internal inductance
energy and inductance linkage and inductance

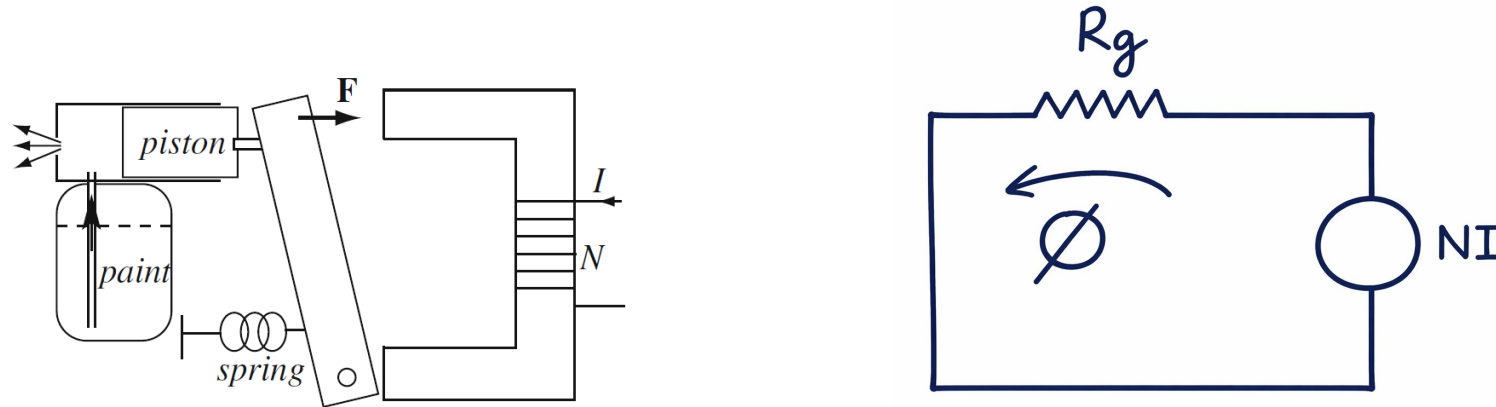
What is still unclear or confusing to you right now? (Responses should reflect your own understanding.)

Derivation of inductance Internal inductance cable inductance
magnetic field **Magnetic energy** external inductance
Energy and Force calculate Dot convention Self inductance
magnetic circuit **magnetic force** mutual flux
virtual displacement **Mutual inductance**
flux linkage
inductance and mutual inductance calculation
equivalent inductance inductance relationships

Magnetic Force calculation from Energy

Magnetic Energy $W_m = \frac{1}{2} LI^2 = \frac{1}{2} \left(\frac{B^2}{\mu} \right) \times \text{Volume}$

Magnetic Force F_m : Gradient of magnetic Energy W_m



Steps:

1. Find B or L
2. Find Energy W_m
3. Find Force by taking the gradient of W_m

For easy understanding, let's consider the iron core of the given structure to be infinitely permeable, i.e., we only have air-gap reluctance. The force causes displacement at the air-gap. Thus, airgap length is a variable (say, x).

Find B

$$\phi = \frac{NI}{\mathcal{R}_g} \rightarrow B.S = \frac{NI}{\mathcal{R}_g}$$

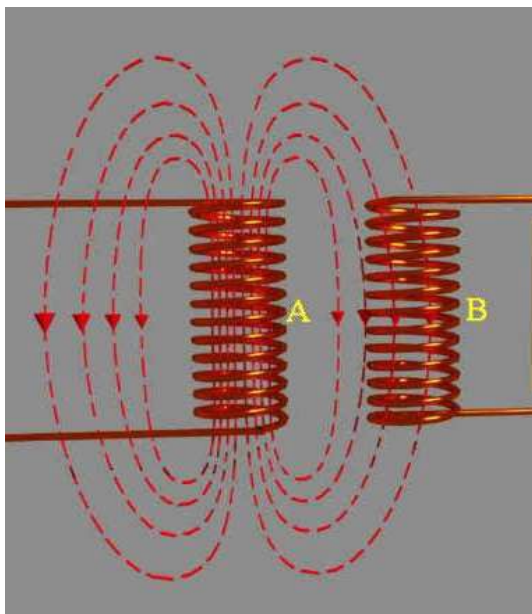
Find Energy W_m

$$W_m = \left(\frac{B^2}{2\mu} \right) \times \text{Volume} = \left(\frac{B^2}{2\mu} \right) \times (S.x)$$

Find Force F_m

$$F_x = \frac{dW_m}{dx} = \frac{d}{dx} \left(\left(\frac{B^2}{2\mu} \right) \times (S.x) \right)$$

Mutual Inductance : Revisit



Mutual inductance is how strongly a changing current in one coil induces a voltage in another coil.

Self-inductance = a coil reacting to its **own magnetic field**

Mutual inductance = a coil reacting to **another coil's magnetic field**

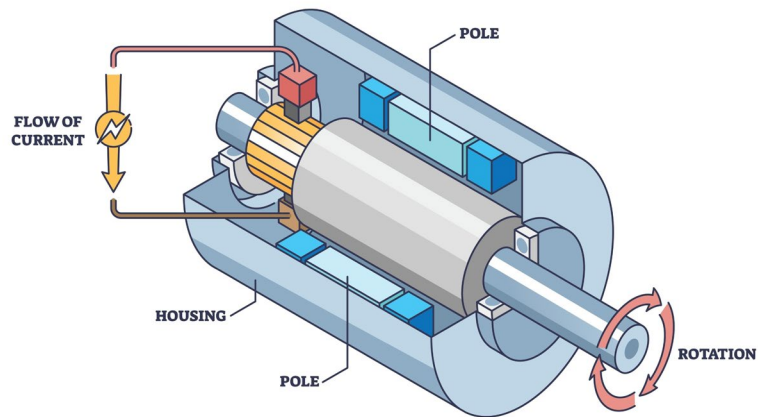
Effect	What changes?	Where voltage appears
Self inductance	Current in coil	Same coil
Mutual inductance	Current in coil 1	Coil 2*

*The dot convention determines whether the voltage of coil 2 is positive or negative at the terminal.

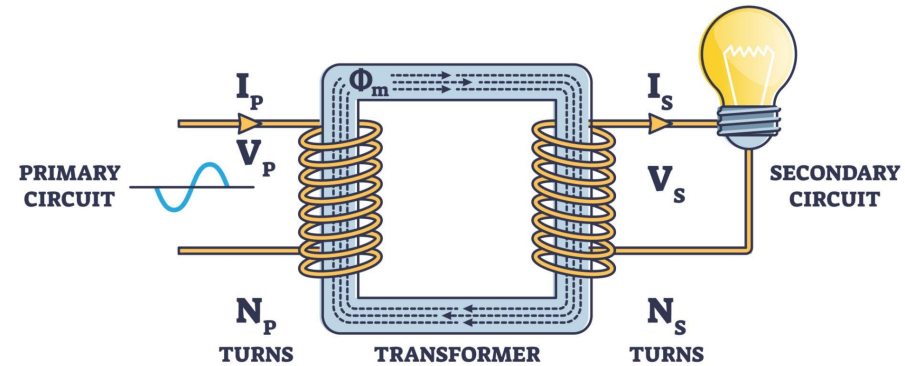
👉 Keep these ideas in mind — today we will extend them.

What is common in these devices?

DC GENERATOR

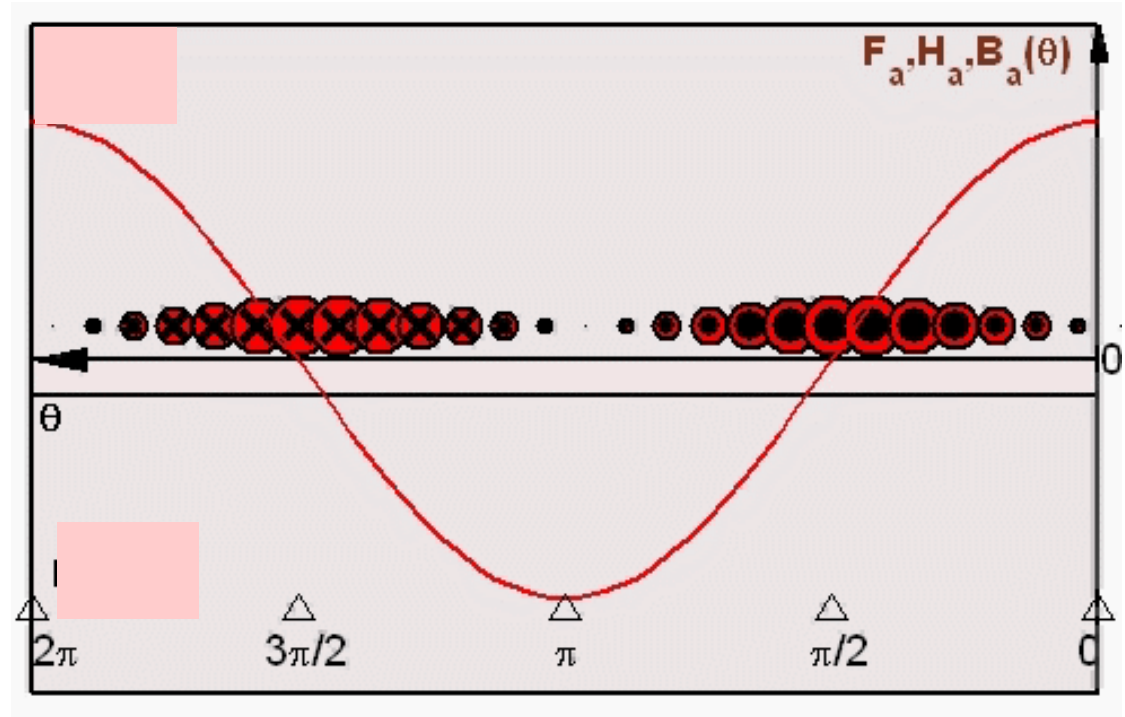


TRANSFORMER

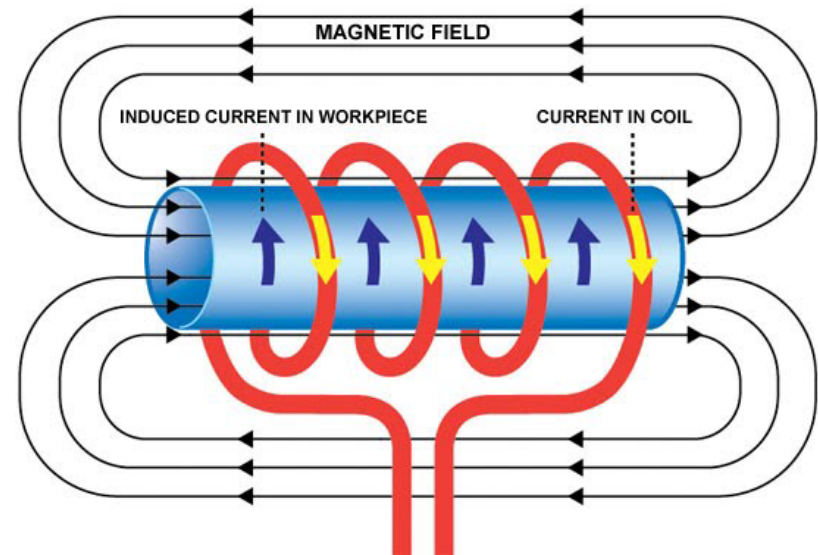


How does the time-varying magnetic field manifest?

What are the consequences of such a variation?



Current induces magnetic field; Can a magnetic field induce current (or voltage)?



Learning goals

After today's lecture, you should be able to



Explain how changing magnetic field create voltage



Apply Faraday's law to magnetic devices (e.g., transformers)



explain Maxwell's equations of electromagnetism
explain eddy current, hysteresis loss and skin effect

Faraday's observation of Electromagnetic induction

Recall Faraday's law:

Voltage is induced in a conducting loop when magnetic flux linking the loop changes with time.

$$e(t) = -N \frac{d\phi}{dt} = -\frac{d\lambda}{dt}$$

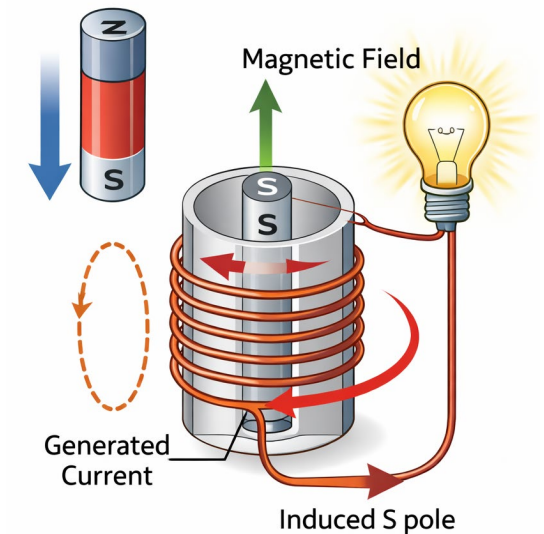
Recall Lenz's law:

The direction of the current due to the induced voltage is such that the flux generated by this current opposes any change in flux.

The induced voltage of Faraday's law pushes current in a closed circuit like other voltage sources, such as a battery.



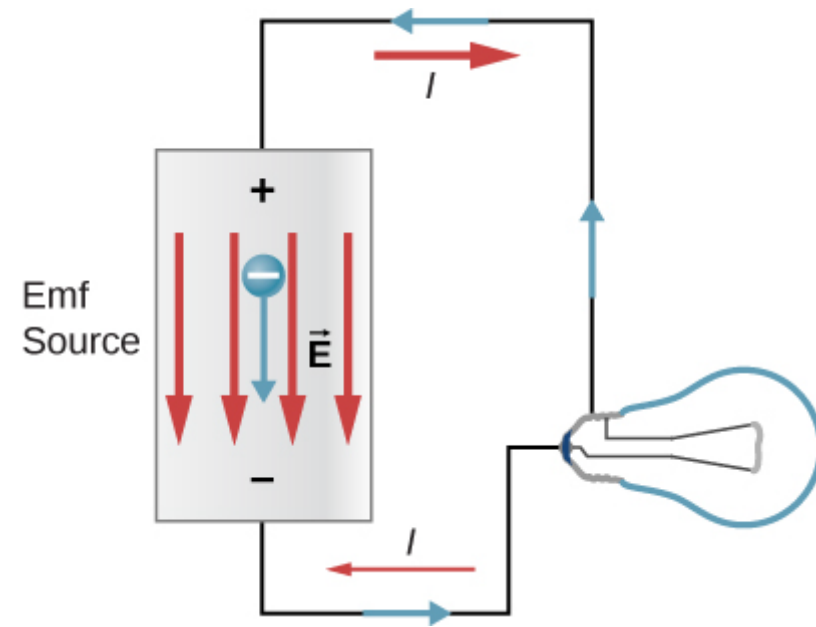
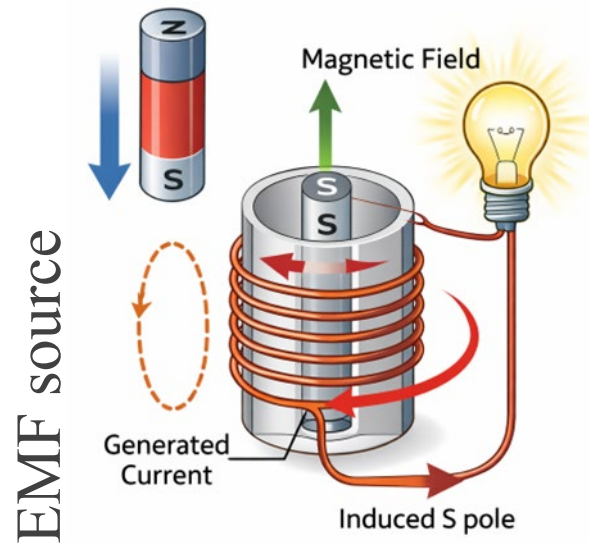
Lenz's Law



EMF (Electromotive Force)

Energy is required to maintain a steady flow of charges (i.e. current) in a closed circuit. The electromotive force (EMF) is the **energy supplied per unit charge to drive current**.

Examples of electrical energy sources: generators, batteries, in which mechanical or chemical energy is converted to electrical energy.

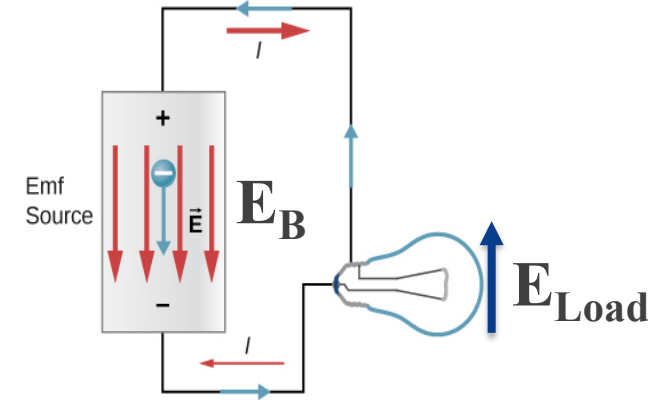


Electric field of an EMF source

The electric field created by charges is conservative: moving a charge **around a closed loop** gives zero net work. $\oint \vec{E} \cdot d\vec{l} = 0$ KVL, $\sum V = 0$

$$\oint \vec{E} \cdot d\vec{l} = 0 \rightarrow \int_B \vec{E}_B \cdot d\vec{l} + \int_{Load} \vec{E}_{Load} \cdot d\vec{l} = 0 \rightarrow \vec{E}_B = -\vec{E}_{Load}$$

$$\oint_C \vec{E}_B \cdot d\vec{l} \neq 0, \text{ instead, } \oint_C \vec{E}_B \cdot d\vec{l} = EMF$$



When we have an EMF (i.e., voltage source), charges are pushed by the source energy (i.e., mech or chem energy): charges are forced to move against the electric field of the source. The charges can gain energy after going around a closed loop when there is an EMF source in the loop. Hence, the electric field associated with the EMF is non-conservative.

Fundamental postulate of electromagnetic induction

Faraday's law: $EMF = -N \frac{d\phi}{dt}$ and $N\phi = \int_s \vec{B} \cdot d\vec{s}$

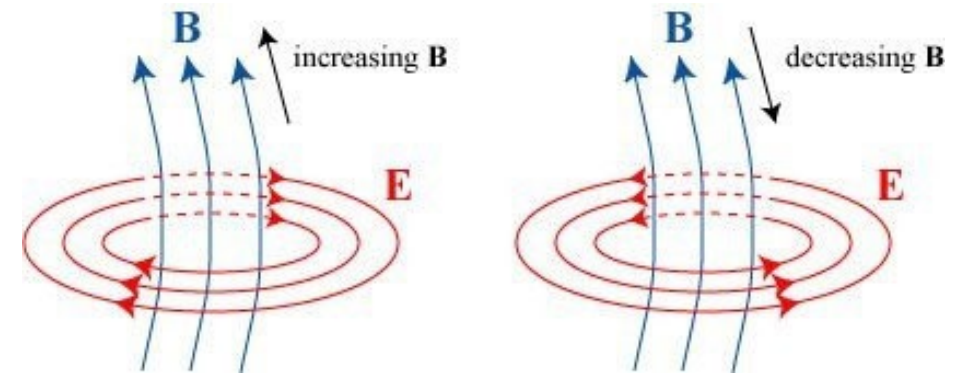
$$\text{Substituting, } \therefore EMF = -\frac{d}{dt} \int_s \vec{B} \cdot d\vec{s} = -\int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} \quad (1)$$

EMF in terms of the associated non-conservative electric field $\mathbf{E}$: $EMF = \oint_C \vec{E} \cdot d\vec{l}$

$$\text{Apply Stokes's theorem: } EMF = \int_s (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} \quad (2)$$

$$\text{From (1) and (2), } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

The curl of $\mathbf{E}$ is zero when $\mathbf{B}$ is non-time-varying – in Electrostatic and Magnetostatic Cases.



The induced electric field $\mathbf{E}$ curls around the time-varying magnetic field $\mathbf{B}$. $\vec{\nabla} \times \vec{E} \neq 0$

Differential form	Integral form
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$EMF = \oint_C \vec{E} \cdot d\vec{l} = -N \frac{d\phi}{dt}$

Important to note that in the presence of a time-varying magnetic field, the curl of $\mathbf{E}$ is not zero. Hence, the negative gradient of EMF (i.e., source voltage V) does not give us the associated non-conservative electric field!!

$$\vec{\nabla} \times \vec{E} \neq 0 \rightarrow \vec{E} \neq -\vec{\nabla} V$$

Displacement current density

Second postulate of magnetostatic: $\vec{\nabla} \times \vec{H} = \vec{J} \rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J}$

From vector property, $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = 0$

But according to continuity equation, $\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$

Div of $\mathbf{J}$ could be zero only if the charge density ρ does not change with time.

This mismatch can be avoided if we consider, $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = 0 = \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t}$,

$$\because \vec{\nabla} \cdot \vec{D} = \rho, \quad \frac{\partial \rho}{\partial t} = \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t}, \quad \vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = 0 \quad \therefore \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$\frac{\partial \vec{D}}{\partial t}$ called **displacement current density**. The mismatch of vector identity can be avoided if we consider the presence of displacement current density in addition to the conduction current density $\mathbf{J}$.

To include the effect of time varying nature of magnetic field, fundamental equations of electric and magnetic field are revised

Electrostatic Magnetostatic	Revised differential form	Revised integral form	Significance
$\vec{\nabla} \times \vec{E} = 0$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_c \vec{E} \cdot d\vec{l} = -N \frac{d\phi}{dt}$	Faraday's Law
$\vec{\nabla} \times \vec{B} = \mu \vec{J}$ $\vec{\nabla} \times \vec{H} = \vec{J}$	$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint_c \vec{H} \cdot d\vec{l} = I + \int_s \frac{\partial \vec{D}}{\partial t} \cdot d\vec{s}$	Ampere's Law (Ampere-Maxwell's law). $\frac{\partial D}{\partial t}$ is negligible in low frequency.
$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon} \rightarrow \vec{\nabla} \cdot \vec{D} = \rho$	-	-	Gauss's law
$\vec{\nabla} \cdot \vec{B} = 0$	-	-	No isolated magnetic pole. (KCL of magnetic flux)



Maxwell's Equations

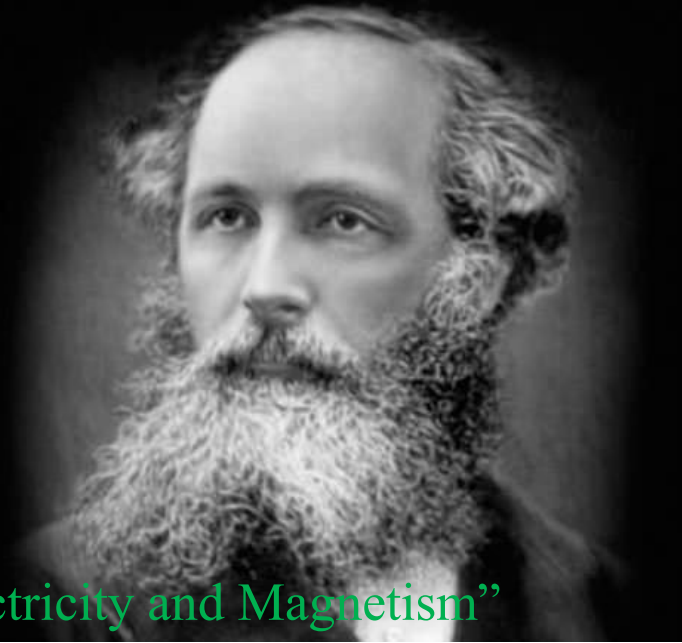
$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

1873 – “Treatise on Electricity and Magnetism”



Significant contribution – concept of displacement current – Ampere-Maxwell's law.

Implications:

1. Interdependence of electric and magnetic fields.
2. Existence of electromagnetic waves.
3. Finite speed of propagation of electromagnetic waves.
4. Propagation in free space is at the speed of light; light itself is an electromagnetic wave.



Einstein's theory of relativity
Existence of gravitational force.

Pause & Reflect

(Before we take a short break...)

 **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

💡 *This is for you — no submission.*

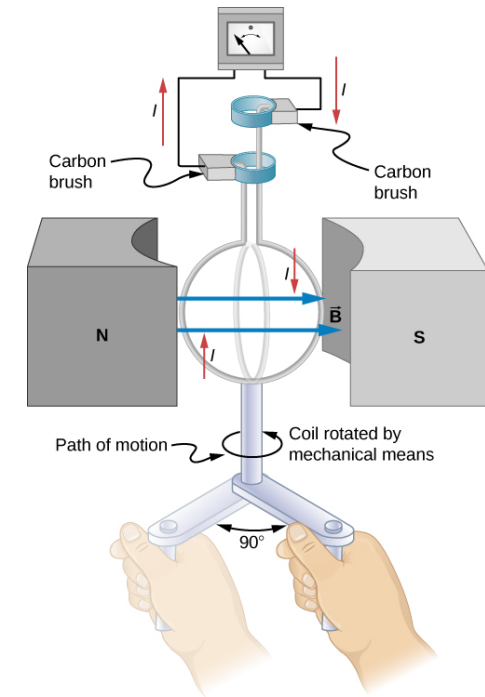
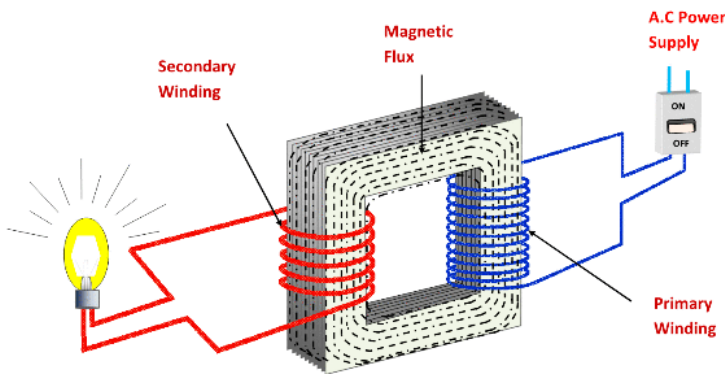


Types of EMFs

$$EMF = -N \frac{d\phi}{dt}, \phi = \int_S \vec{B} \cdot d\vec{s} \quad (\text{Faraday's law of induction})$$

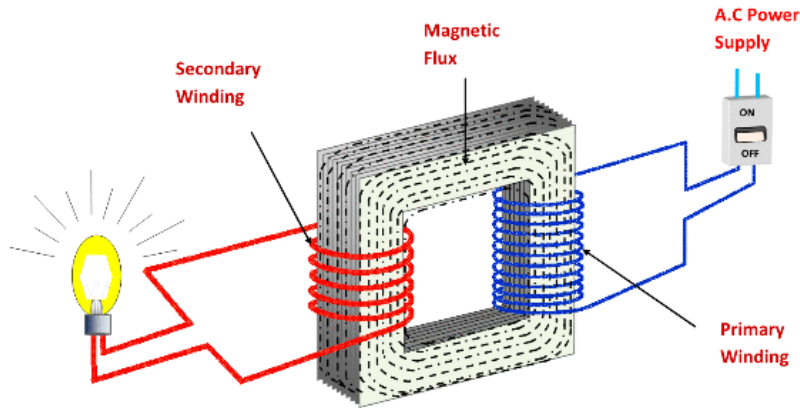
Three possible scenarios for the induced emf:

1. **Transformer EMF:** time-varying magnetic field and stationary conductors, i.e., $\phi(t)$
2. **Motional EMF:** Conductor moving in the magnetic field to produce a $d\phi/dt$
3. **Combined effect:** moving conductor and changing field



Transformer action

The circuit is stationary, and the magnetic field is time-varying. The induced EMF is *due to transformer action*. Example – transformers



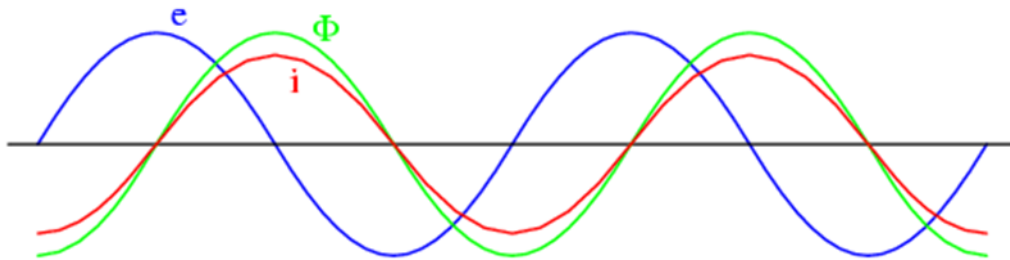
$$EMF = -N \frac{d\phi}{dt}, \phi = \int_s \vec{B}(t) \cdot d\vec{s}$$

where, $B(t) = B_m \cos \omega t \leftarrow i(t) = I_m \cos \omega t$

Assuming $B(t)$ is uniform in the core:

$$\therefore \phi(t) = B(t)S = B_m S \cos \omega t = \phi_m \cos \omega t$$

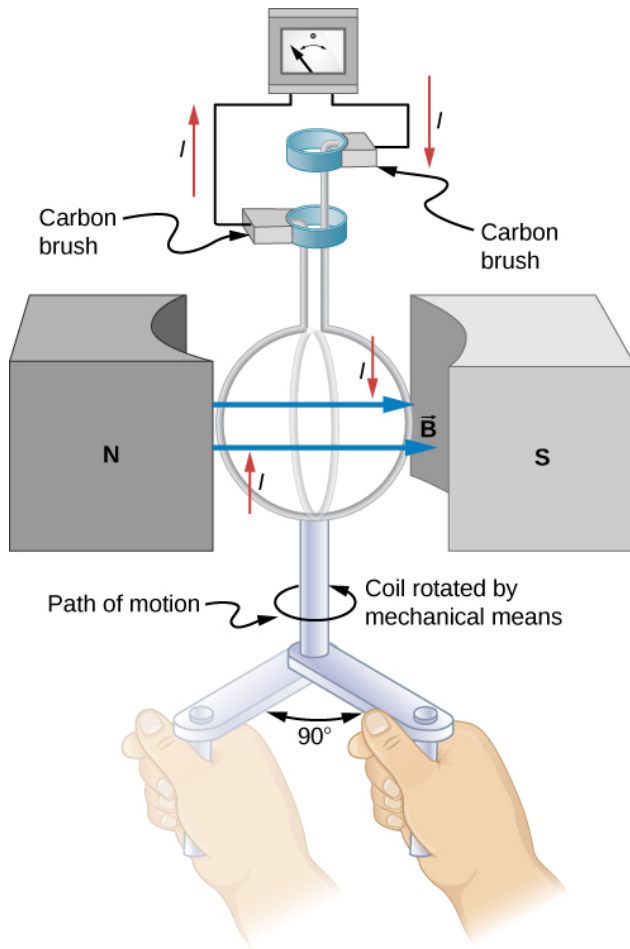
e = voltage Φ = magnetic flux i = coil current



$$e(t) = -N \frac{d\phi(t)}{dt} = (N\phi_m\omega) \sin \omega t = E_m \sin \omega t$$

Motional EMF

A current-carrying conductor (circuit) moving with velocity $\mathbf{u}$ in the static magnetic field $\mathbf{B}$ induces *Motional EMF*.
Example –generators with DC field excitation.



$$EMF = \oint_C \vec{E} \cdot d\vec{l}$$

Lorenz's force: $\vec{F} = q(\vec{u} \times \vec{B}) \rightarrow \frac{\vec{F}}{q} = \vec{u} \times \vec{B} = \vec{E}$

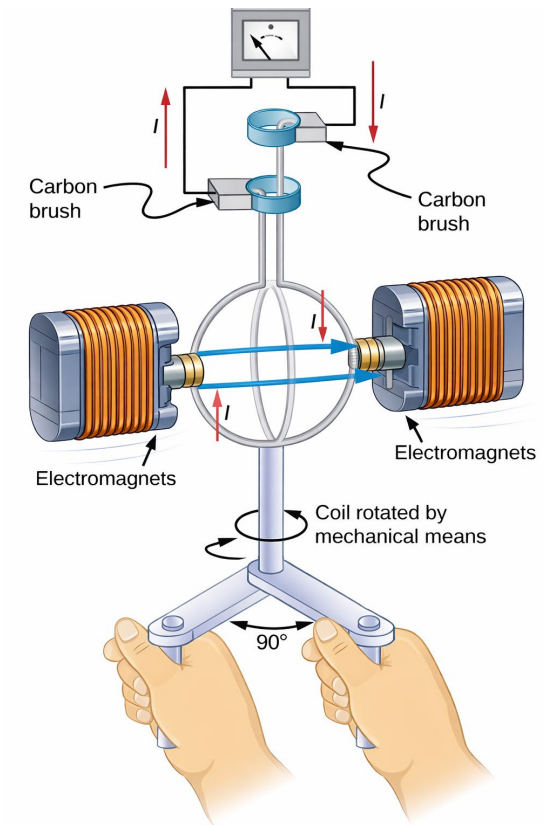
$$\therefore EMF = \oint_C \vec{E} \cdot d\vec{l} = \oint_C (\vec{u} \times \vec{B}) \cdot d\vec{l}$$

Combined Effect : EMF

A moving current-carrying conductor in a time-varying magnetic field $\mathbf{B}(t)$ induces combined *transformer action and motional emf*. Example – Generators with AC field excitation.

EMF = Motional EMF + Transformer EMF

$$EMF = \oint_C (\vec{u} \times \vec{B}(t)) \cdot d\vec{l} - N \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

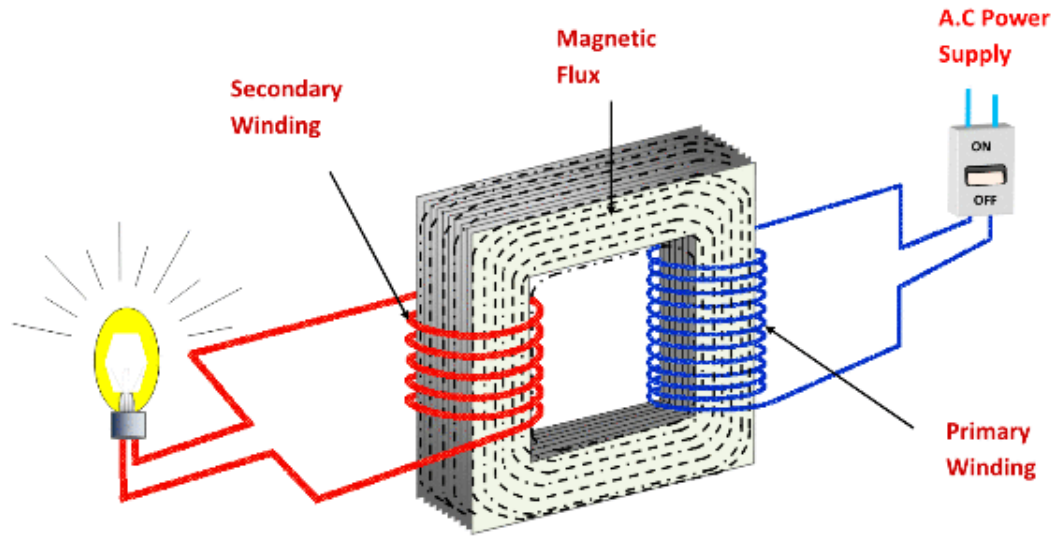


Transformers



One of the most important devices that enables highly efficient electrical energy transfer. Used not just in high power transmission but also in electronics and instrumentation for isolation and measurement purposes.

AC current $\rightarrow$ time varying flux $\rightarrow$ induced voltage



AC current in the primary winding i_p :

$$i_p(t) = I_m \cos \omega t,$$

Ampere's law
 $(\oint \vec{B} \cdot d\vec{l} = \mu N_p i_p)$



$$B(t) = B_m \cos \omega t \rightarrow \Phi(t) = \Phi_m \cos \omega t$$

$$\left(B_m = \frac{\mu N_p I_m}{l_c} \right) \quad (\phi_m = B_m S_c)$$

By Faraday's law:

$$e_p(t) = -N_p \frac{d\phi(t)}{dt} = N_p \omega \Phi_m \sin \omega t = E_{pk1} \sin \omega t$$

$$e_s(t) = -N_s \frac{d\phi(t)}{dt} = N_s \omega \Phi_m \sin \omega t = E_{pk2} \sin \omega t$$



In an ideal transformer, terminal voltages and induced EMFs are the same because we ignore the resistance and inductance of the two coils.

$$v_p(t) = e_p(t) = N_p \omega \Phi_m \sin \omega t = E_{pk1} \sin \omega t$$

$$v_s(t) = e_s(t) = N_s \omega \Phi_m \sin \omega t = E_{pk2} \sin \omega t$$

$$\omega = 2\pi f$$

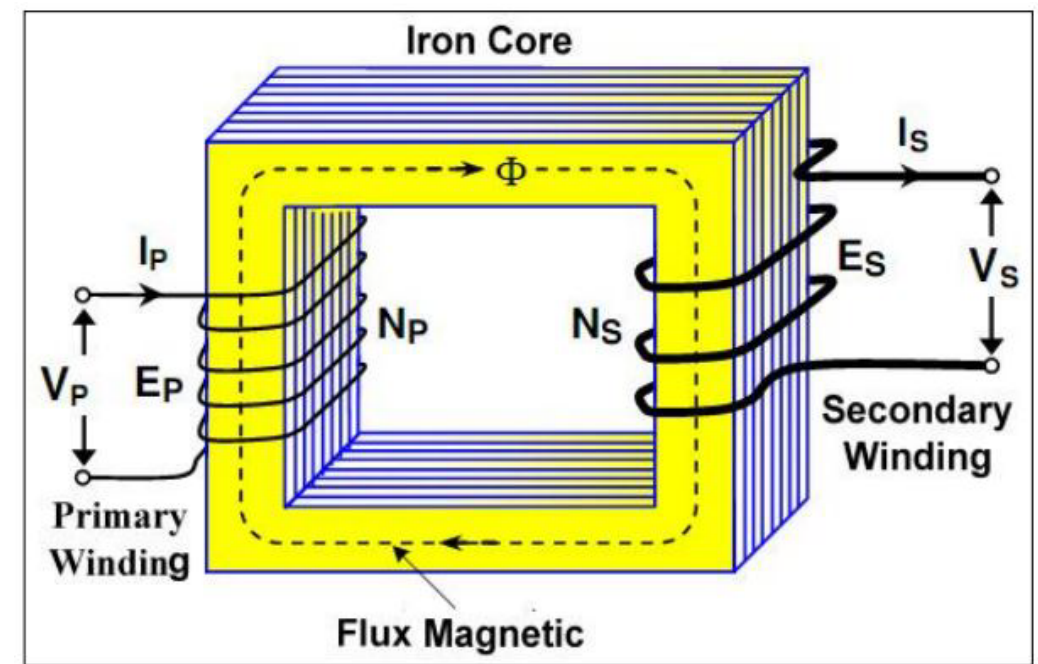
RMS value of the primary side voltage:

$$V_p = \frac{E_{pk1}}{\sqrt{2}} = \frac{\omega N_p \phi_m}{\sqrt{2}} = \frac{2\pi f N_p \phi_m}{\sqrt{2}} = 4.44 f N_p \phi_m = 4.44 f N_p (B_m S_c)$$

Similarly, the RMS value of the secondary side voltage, $V_s = 4.44 N_s f \phi_m$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S}$$

Can you draw the *magnetic circuit* and show that $N_p/N_s = I_s/I_p$?
You may consider the core to be infinitely permeable.

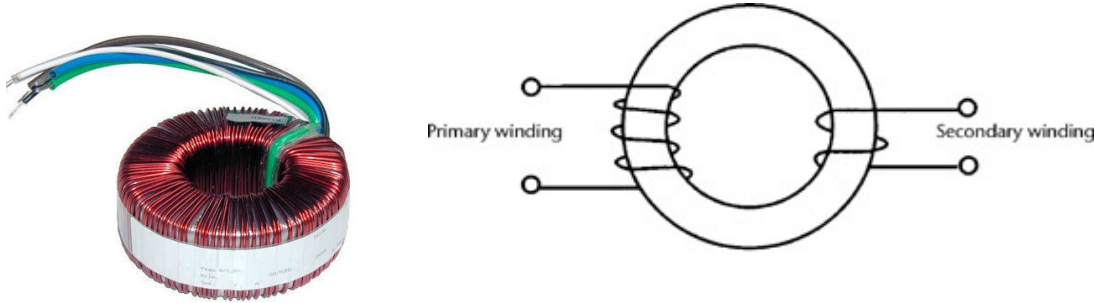


Example 8

A widely used transformer in high-quality audio and test equipment is built around a toroidal core, as shown in Fig.

(a) The primary coil consists of $N_1 = 1500$ turns and carries an AC current $I = I_m \sin \omega t$. The mean radius of the toroid core is $r = 125 \text{ mm}$, cross-sectional area $S = 1000 \text{ mm}^2$. The relative permeability of the core material is 3000. Find the peak value of current I_m required to induce an EMF 240 V (rms) in the primary winding. Consider the frequency of the AC current is 50 Hz, and assume the flux density in the core is uniform. The secondary winding is an open circuit. **Ans: 0.1 A**

(b) If the transformer is required to connect to an AC supply of frequency 16.67 Hz, what constraints, if any, must be imposed on the use of this transformer? Consider that the supply voltage can be adjusted.



Solution steps:

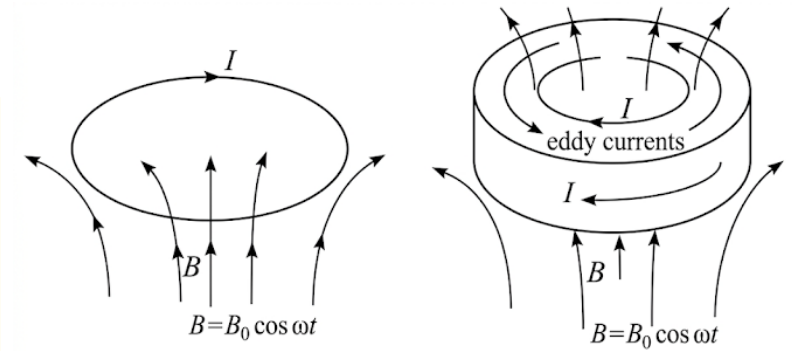
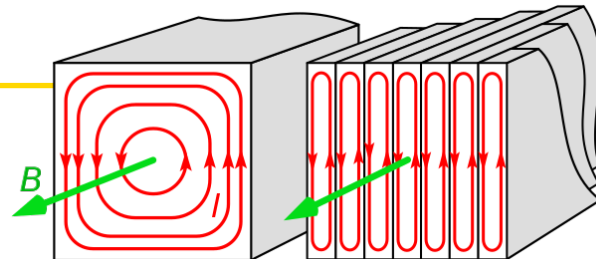
1. Find ϕ_m from $V_1 = 4.44 f N_1 \phi_m$
2. Find I_m from $\phi_m = \frac{N_1 I_m}{\mathfrak{R}}$, $\mathfrak{R} = \frac{l_c}{\mu S}$

Eddy current and losses

When a conducting loop is in a time-varying magnetic field B , an EMF and hence a current is induced in the loop by Faraday's law. A similar current is also induced if a conducting material is placed in B .

The induced circulating currents in the conducting material due to the time-varying field are called '*Eddy currents*'. Like any other conduction current, eddy current generates heat losses ($I_{Eddy}^2 R$ or emf^2/R) known as **Eddy current loss**.

Instead of a solid piece, if the conductor is made with laminated sheets electrically insulated from each other to break up the path of eddy current, heat loss will be minimal.



Cause

Time-varying magnetic field



Effect

Circulating current



Result

Heat loss



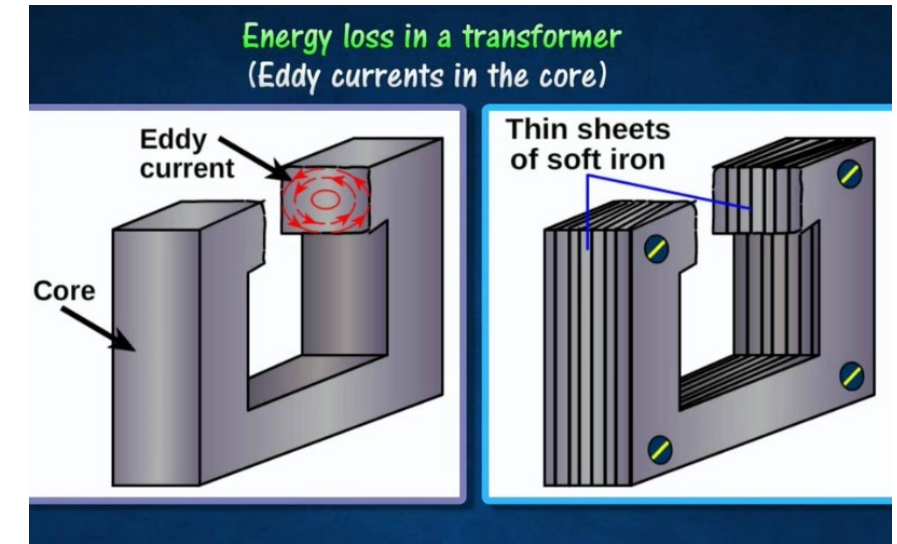
Solution

Laminated sheets



Eddy current loss in magnetic cores:

Most magnetic materials are also good conductors with a conductivity σ . Heat loss due to eddy current can occur in the core of the magnetic devices, such as a transformer. The eddy current loss in the magnetic core is undesirable. Hence, magnetic cores are made with a number of thin lamination sheets stacked together.

$$B(t) = B_m \cos \omega t \rightarrow \phi(t) = \phi_m \cos \omega t,$$


$$\text{Induced EMF: } \oint_C \vec{E} \cdot d\vec{l} = -\frac{d\phi(t)}{dt}, \quad \text{Induced eddy current density: } \vec{J}_{eddy} = \sigma \vec{E}$$

$$\text{Power loss per unit volume: } p_e = \vec{E} \bullet \vec{J}_{eddy} = \frac{|\vec{J}_{eddy}|^2}{\sigma} = \sigma |\vec{E}|^2 \quad [\text{W/m}^3]$$

Power loss per unit volume of a laminated core with lamination thickness τ and for an AC supply frequency f Hz:

$$p_e = k_e f^2 B_m^2 \quad (\text{W/m}^3), \quad k_e = \frac{\pi^2 \sigma}{6} \tau \quad (k_e: \text{eddy current loss coefficient})$$


Application Example 9

A thin ring of mean radius $r = 10$ mm and cross-sectional area $S = 1$ mm² is placed on a time-varying magnetic field $B(t) = (1 + 60t)$ Tesla as shown in the figure. Due to the time-variation of the flux density, the eddy current induces in the ring. The conductivity of the ring is $\sigma = 10^7$ S/m. Calculate the magnitude of the eddy current and the heat loss (i.e. I^2R). **Ans: 3A**

$$EMF = \oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}, \quad \vec{J}_{eddy} = \sigma \vec{E} \rightarrow I_{eddy} = \frac{EMF}{R}$$

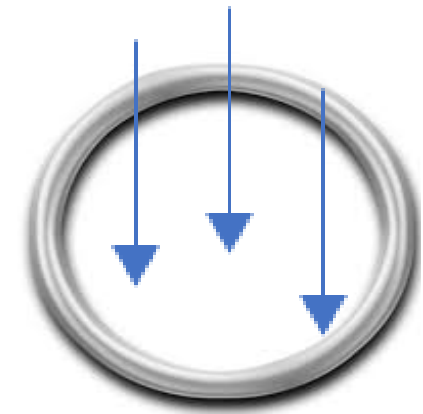
Solution steps

Find EMF in the ring: $EMF = - \frac{d\phi(t)}{dt}, \phi(t) = B(t) \cdot S$

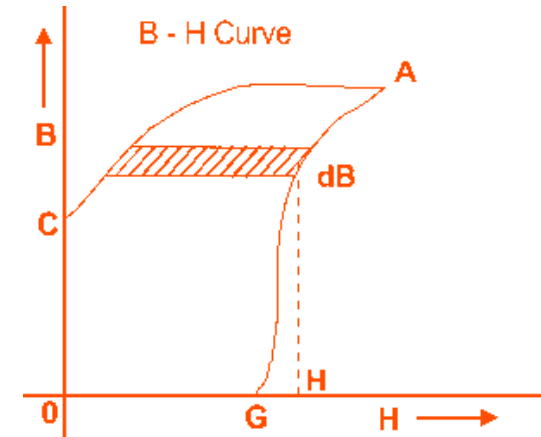
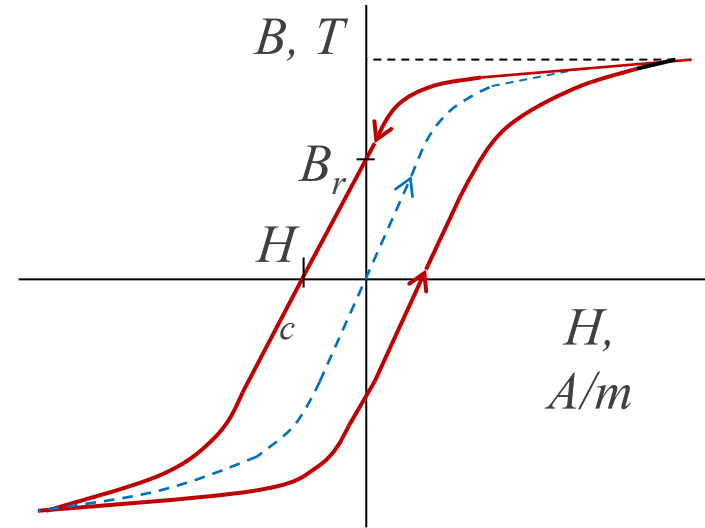
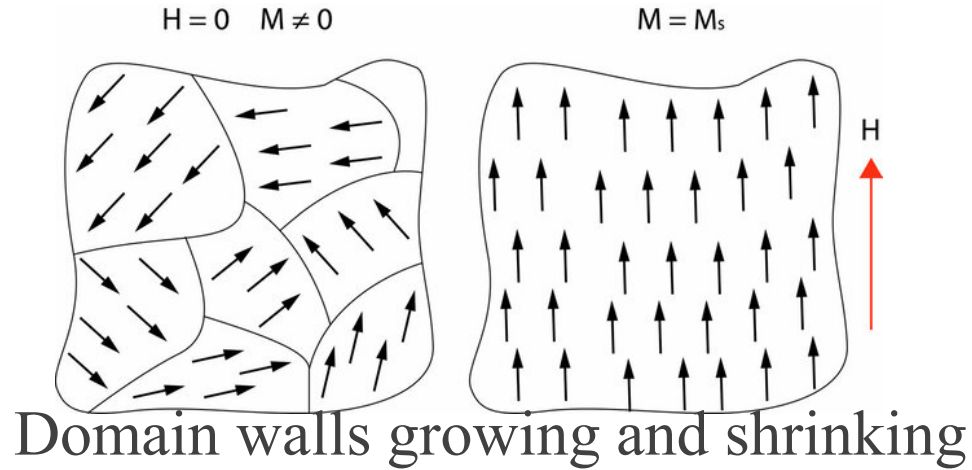
Find resistance of the Ring $R = \frac{l}{\sigma S}$

Find eddy current $I_{eddy} = \frac{EMF}{R}$

Find loss $I_{eddy}^2 R$



Hysteresis loss



Recall the hysteresis loop of magnetic material. With ac excitation, the magnetization and demagnetization of the core material happens repeatedly. The process is related to the growing and shrinking of domain walls inside the magnetic material. Repeated growing and shrinking of domain walls results in heat loss of the core and known as **hysteresis loss**.

It can be shown experimentally that **energy loss per cycle = area of B-H loop** = KB_m^n

Hysteresis loss per unit volume can also be found as, $p_h = K_h f B_m^n$ (W/m³)

where, K_h : Hysteresis coefficient, f : frequency in Hz, B_m : Maximum flux density in the core in T,
 n : Steinmetz constant, the value of n varies between 1.6 to 2.



Core or Iron loss

Eddy current loss and hysteresis loss are often grouped together and known as iron or core loss, which occurs in the core of the magnetic devices.

Core loss density in Watts per m^3 :

$$p_c = p_h + p_e = K_h f B_m^n + K_e f^2 B_m^2 \left[W/m^3 \right]$$

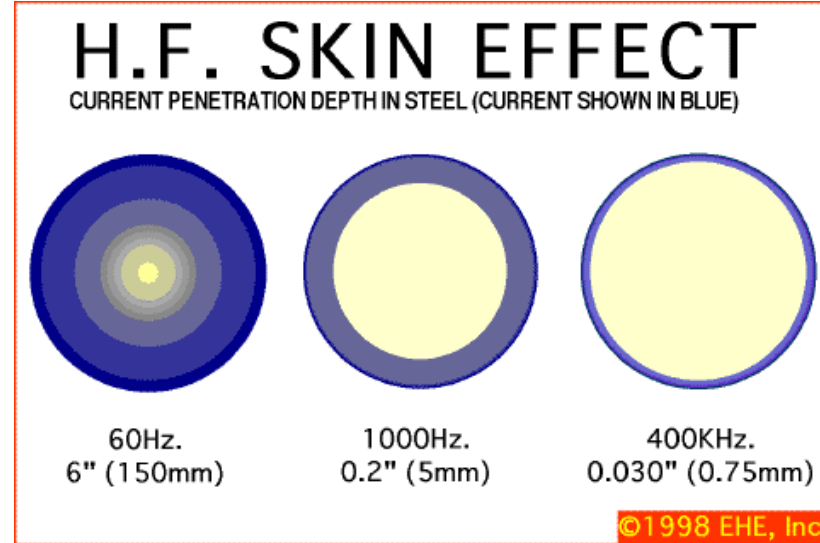
Thus, the core loss depends on frequency f and peak magnetic field B_m in the core. The larger the frequency f and field B_m , the higher the core loss in a device.

Apart from core or iron loss, we may also have losses in the resistance of the coils. (referred to as copper loss).



Skin effect

When the electric field $\mathbf{E}$ varies rapidly (high frequency) in a conductor, the current inside the conductor becomes non-uniform. At a sufficiently high frequency, the interior region of the conductor may not carry any current, and all current may crowd at the surface – Skin effect



High-frequency current
→ changing magnetic field inside conductor
→ induced currents oppose interior current
→ current crowds to surface.

Skin depth:

$$\begin{aligned}\delta &= \sqrt{\frac{2}{\sigma\mu\omega}} \\ &= \sqrt{\frac{2}{\sigma\mu(2\pi f)}} = \sqrt{\frac{1}{\sigma\mu(\pi f)}}\end{aligned}$$



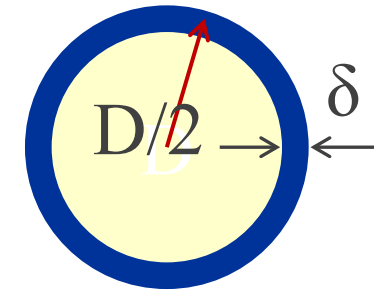
AC Resistance due to skin effect

A conductor carrying an AC current has a reduced effective cross-sectional area S because of the **skin effect**. Hence, the conductors carrying AC current have a higher resistance than they would have when carrying DC current.

AC resistance of a conductor of length l and diameter D is approximately,

$$R_{dc} = \frac{l}{\sigma S} \quad (\text{Ohm}), \quad S = \pi \left(\frac{D}{2} \right)^2$$

$$R_{ac} = \frac{1}{\sigma} \left(\frac{l}{2\pi \frac{(D-\delta)}{2} \delta} \right) \approx \frac{1}{\delta \sigma} \left(\frac{l}{\pi D} \right) \quad (\text{Ohm})$$



AC resistance need to be considered when the circuit operates at high frequency.

Example 10

A copper conductor of circular cross section (see figure) carries current at 10 kHz . The diameter of the conductor is 5 mm. Assuming that nearly all of the resistive (i.e., I^2R) power loss in the conductor takes place within the skin depth (δ), find the resistance per meter of the conductor for currents at frequencies of 10 kHz.

Express the % increase in resistance of the conductor compared to the resistance of the conductor when DC current flows, i.e., at 0 Hz.

Permeability $\mu = 4\pi \times 10^{-7}$ H/m, Conductivity $\sigma = 5 \times 10^7$ S/m

Solution steps:

1. Compute skin depth $\delta = \sqrt{\frac{1}{\sigma\mu(\pi f)}} = 0.71\text{mm}$

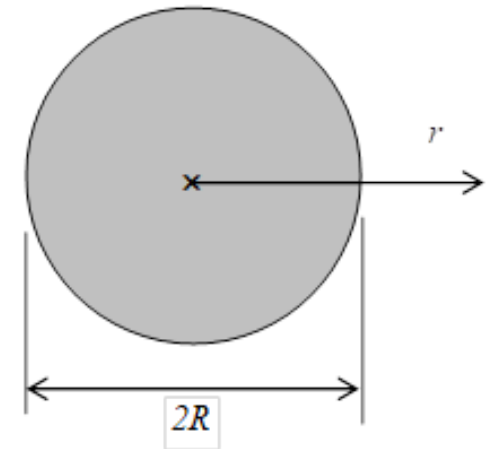
2. Determine the effective areas

3. Calculate AC and DC resistances

$$R_{ac} \approx \frac{1}{\delta\sigma} \left(\frac{l}{\pi D} \right) \text{ (Ohm)} = 1.8\text{m}\Omega/\text{m}$$

$$R_{dc} = \frac{l}{\sigma S} \text{ (Ohm)}, S = \pi \left(\frac{D}{2} \right)^2 = 1 \text{ m}\Omega/\text{m}$$

4. Compare the two resistances



Does it mean DC circuits are more efficient?

Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.



0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud

All responses

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1 of 3

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Scan the QR or use link to join



<https://forms.office.com/r/vJs4eSWdQH>

 Copy link

Key Takeaways

- ✓ Changing magnetic fields produce voltage
- ✓ Maxwell unified electric and magnetic fields
- ✓ Transformers use electromagnetic induction
- ✓ Eddy current and hysteresis cause core losses
- ✓ High frequency currents create skin effect

More examples for review (recommended)

Example 7-1, 7-4, 7-5 of your textbook by D.K. Cheng.

End of low frequency Electromagnetics- Part A.

